

Pemrograman Web – BIE02

Intro to PHP

Sendy Ferdian Sujadi

PHP

- PHP : Hypertext Preprocessor
 - Kode ditanamkan pada HTML.
 - Server side scripting
 - Butuh web server, ex: Apache Web Server
 - Open source and **FREE**
 - Dapat digunakan pada hampir semua sistem operasi

More on PHP

- PHP script starts with
`<?php //write code ?>`
- PHP script can be put anywhere in a document
- PHP file has .php for extension
- A PHP files can contain html tags
- Almost every line of php needs a semicolon very much like a programming language

Instalasi PHP

- XAMPP?
 - Alasannya mudah dan sudah mencakup beberapa paket penting yaitu:
 - PHP library,
 - web server (Apache HTTP Server),
 - basis data: MySQL (dengan aplikasi pengelola database MySQL : phpmyadmin)
- Text Editor: Notepad / Notepad++ / VSCode / dll
- HTML Editor yang canggih. (ex: Macromedia / Adobe Dreamweaver)

Memulai PHP → First PHP

1. Jalankan service Apache.
2. Buat direktori dalam web server, dengan path direktori :
C:\xampp\htdocs\ dengan nama direktori = “**maranatha**”
3. Buka texteditor dan ketikkan : (simpan dengan nama **index.php**)

```
<html>  
  <body>  
    <?php  
      echo "Hello World";  
    ?>  
  </body>  
</html>
```
4. Buka browser kemudian ketik alamat “<http://localhost/maranatha>”

Dasar-dasar PHP

- Embedded Script
 - File php ditanamkan / ditempelkan pada tag HTML
- Non-Embedded Script
 - Langsung dengan tag PHP

```
contoh_php_embeddedScript.php
1  <!DOCTYPE html>
2  <html>
3
4  <head>
5  |   <title>Contoh Embedded Script</title>
6  </head>
7
8  <body>
9  |   <a href="index.html">home</a></br>
10 |   <?php
11 |   //Contoh PHP Embedded Script
12 |   echo "Hello World!";
13 |   ?>
14 </body>
15
16 </html>
```

```
contoh_php_nonEmbeddedScript.php
1  <!DOCTYPE html>
2  <?php
3  echo("<html>");
4  echo("<head>");
5  echo("<title>Contoh Non Embedded Script</title>");
6  echo("</head>");
7  echo "<body bgcolor= \"blue\">";
8  echo "<a href='index.html'>home</a></br>";
9  echo "Hello World!";
10 echo "</body>";
11 echo "</html>";
12 ?>
```

Variable dan Tipe data (1/2)

- Nama Variabel di PHP harus diawali dengan tanda \$ dan diikuti nama variable, cth: \$total
- PHP akan melakukan konversi data, sesuai dengan VALUE yang di ASSIGN dalam variable untuk masing-masing variable seperti:
 - Tipe data Boolean: true | false
 - Tipe data integer: 123
 - Tipe data float: 123.12
 - Tipe data string: "Andi" | 'Budi'
 - Tipe data string ke integer
- Konstanta:
 - `Define ("NAMA_KONSTANTA", nilai_konstanta);`

Variable dan Tipe data (2/2)

PHPVariable.php > ...

```
1 <html>
2
3 <head>
4   <title>PHP Variable</title>
5 </head>
6
7 <body>
8   <?php
9     Define ("pajak", 0.11);
10    $jumlah = '5';
11    $harga = 20000;
12    $pajak = ($harga * $jumlah) * pajak;
13    $total = $harga * $jumlah + $pajak;
14    echo "Jumlah Beli : $jumlah <br>";
15    echo "Harga Barang : $harga <br>";
16    echo "Pajak : $pajak <br>";
17    echo "Total Bayar : $total <br>";
18  ?>
19 </body>
20
21 </html>
```

Jumlah Beli : 5

Harga Barang : 20000

Pajak : 11000

Total Bayar : 111000

Komentar pada PHP

🐘 komentar.php > ...

```
1  <?php
2      //konstanta untuk nilai judul
3      define("Judul", "Hitung Luas Lingkaran");
4      #konstanta untuk nilai phi
5      define("PHI", 3.14);
6
7      echo "<html>";
8      echo "<head>";
9      echo "<title>" . Judul . "</title>";
10     echo "</head>";
11     echo "<body>";
12     echo "<a href='index.html'>home</a></br>";
13     echo "Jari-jari : " . ($r=5) . " <br>";
14     /**
15      * Rumus Luas Lingkaran:
16      *  PHI * Jari-Jari * Jari-Jari
17      */
18     $luas = PHI * $r * $r;
19     echo "Luas Lingkaran = $luas";
20     echo "</body>";
21     echo "</html>";
22     ?>
```

PHP operator (1/4)

- Arithmetic operator

Operator	Name	Description	Example	Result
$x + y$	Addition	Sum of x and y	$2 + 2$	4
$x - y$	Subtraction	Difference of x and y	$5 - 2$	3
$x * y$	Multiplication	Product of x and y	$5 * 2$	10
x / y	Division	Quotient of x and y	$15 / 5$	3
$x \% y$	Modulus	Remainder of x divided by y	$5 \% 2$ $10 \% 8$ $10 \% 2$	1 2 0
$- x$	Negation	Opposite of x	$- 2$	
$a . b$	Concatenation	Concatenate two strings	"Hi" . "Ha"	HiHa

PHP operator (2/4)

- Assignment operator

Assignment	Same as...	Description
<code>x = y</code>	<code>x = y</code>	The left operand gets set to the value of the expression on the right
<code>x += y</code>	<code>x = x + y</code>	Addition
<code>x -= y</code>	<code>x = x - y</code>	Subtraction
<code>x *= y</code>	<code>x = x * y</code>	Multiplication
<code>x /= y</code>	<code>x = x / y</code>	Division
<code>x %= y</code>	<code>x = x % y</code>	Modulus
<code>a .= b</code>	<code>a = a . b</code>	Concatenate two strings

PHP Operator (3/4)

- Boolean operator - Comparison

Operator	Name	Description	Example
<code>x == y</code>	Equal	True if x is equal to y	<code>5==8</code> returns false
<code>x === y</code>	Identical	True if x is equal to y, and they are of same type	<code>5=== "5"</code> returns false
<code>x != y</code>	Not equal	True if x is not equal to y	<code>5!=8</code> returns true
<code>x <> y</code>	Not equal	True if x is not equal to y	<code>5<>8</code> returns true
<code>x !== y</code>	Not identical	True if x is not equal to y, or they are not of same type	<code>5!== "5"</code> returns true
<code>x > y</code>	Greater than	True if x is greater than y	<code>5>8</code> returns false
<code>x < y</code>	Less than	True if x is less than y	<code>5<8</code> returns true
<code>x >= y</code>	Greater than or equal to	True if x is greater than or equal to y	<code>5>=8</code> returns false
<code>x <= y</code>	Less than or equal to	True if x is less than or equal to y	<code>5<=8</code> returns true

PHP operator (4/4)

- Boolean operator(2) - logic

Operator	Name	Description	Example
x and y	And	True if both x and y are true	x=6 y=3 (x < 10 and y > 1) returns true
x or y	Or	True if either or both x and y are true	x=6 y=3 (x==6 or y==5) returns true
x xor y	Xor	True if either x or y is true, but not both	x=6 y=3 (x==6 xor y==3) returns false
x && y	And	True if both x and y are true	x=6 y=3 (x < 10 && y > 1) returns true
x y	Or	True if either or both x and y are true	x=6 y=3 (x==5 y==5) returns false
! x	Not	True if x is not true	x=6 y=3 !(x==y) returns true

Getting value from GET

- GET akan mengirimkan data bersama dengan URL
- Use `$_GET["x"]`; with x as the name of your input type

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Masukkan nama :

Form Metode Get Proses

x

+

localhost/maranatha/metodegetproses.php?nama=Galaxy+Bima+Sakti

[home](#)

Data Nama Yang Diinputkan adalah : Galaxy Bima Sakti

Getting value from POST

- POST akan mengirimkannya secara terpisah
- Use `$_POST["x"]`; with x as the name of your input type

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Masukkan nama :

Form Metode post Proses

localhost/maranatha/metodepostproses.php

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Data Nama Yang Diinputkan adalah : Galaxy Bima Sakti

Example Case GET and POST

```
1 <html>
2
3 <head>
4 |   <title> Form Metode Get</title>
5 </head>
6
7 <body>
8 |   <a href="index.html">home</a></br>
9 |   <form action="metode....proses.php" method="....">
10 |     Masukkan nama : <input type="text" name="nama" size="25">
11 |     <input type="submit" value="Proses">
12 |   </form>
13 </body>
14
15 </html>
```

.... diubah dengan get | post

🐾 metodegetproses.php

```
1 <html>
2
3 <head>
4 |   <title> Form Metode Get Proses</title>
5 </head>
6
7 <body>
8 |   <a href="index.html">home</a></br>
9 |   Data Nama Yang Diinputkan adalah :
10 |   <?php echo $_GET["nama"]; ?>
11 </body>
12
13 </html>
```

🐾 metodepostproses.php

```
1 <html>
2
3 <head>
4 |   <title> Form Metode post Proses</title>
5 </head>
6
7 <body>
8 |   <a href="index.html">home</a></br>
9 |   Data Nama Yang Diinputkan adalah :
10 |   <?php echo $_POST["nama"]; ?>
11 </body>
12
13 </html>
```


Form Input dan Output dalam 1 halaman

- Perlu diingat tag `<form>` untuk aksi diperlukan 2 atribut penting:
 - action: akan memanggil file .php sebagai proses
 - method: GET atau POST
- Agar halaman dapat mengakses action ke diri sendiri bisa menggunakan script php `"$_server['php_self']"`

```
SelfForm.php
18 <html>
19
20 <head>
21   <title>PHP Self Form</title>
22 </head>
23
24 <body>
25   <a href="index.html">home</a></br>
26   <h1> Masukan Identitas Anda</h1>
27   <form action="<?php echo $_SERVER['PHP_SELF']; ?>" method="post">
28     Isikan Nama : <input type="text" name="nama" /></br>
29     Isikan No Telp: <input type="text" name="notelp" /></br>
30     Isikan Alamat : <textarea name="alamat" rows="5" cols="40"></textarea></br>
31     <input type="submit" value="TAMPIL"><input type="reset" value="BATAL" /></br>
32   </form>
33   <?php
34     $nama = $_POST['nama'];
35     $telp = $_POST['notelp'];
36     $alamat = $_POST['alamat'];
37     if (!empty($nama)) {
38       echo "Nama : $nama <br>";
39     }
40     if (!empty($telp)) {
41       echo "No Telp : $telp <br>";
42     }
43     if (!empty($alamat)) {
44       echo "Alamat : $alamat <br>";
45     }
46     ?>
47   </body>
48
49 </html>
```

PHP Percabangan

```
<?php
$nilai = 40;
if ($nilai >= 60) {
    echo "Nilai Anda = $nilai. Selamat, Anda Lulus";
} else {
    echo "Nilai Anda = $nilai. Sorry, Anda Tidak Lulus";
}
?>
```

1

```
<?php
$nilai = 90;
if (($nilai >= 0) && ($nilai < 50)) {
    $grade = "E";
} elseif (($nilai >= 50) && ($nilai < 60)) {
    $grade = "D";
} elseif (($nilai >= 60) && ($nilai < 75)) {
    $grade = "C";
} elseif (($nilai >= 75) && ($nilai < 85)) {
    $grade = "B";
} elseif (($nilai >= 85) && ($nilai < 100)) {
    $grade = "A";
} else {
    $grade = "Nilai anda di luar jangkauan";
}
echo "Nilai Anda : $nilai, dikonversi menjadi $grade";
?>
```

2

```
<?php
$angka = 6;
switch ($angka) {
    case 0:
        $terbilang = "NOL";
        break;
    case 1:
        $terbilang = "SATU";
        break;
    case 2:
        $terbilang = "DUA";
        break;
    case 3:
        $terbilang = "TIGA";
        break;
    case 4:
        $terbilang = "EMPAT";
        break;
    case 5:
        $terbilang = "LIMA";
        break;
    case 6:
        $terbilang = "ENAM";
        break;
    case 7:
        $terbilang = "TUJUH";
        break;
    case 8:
        $terbilang = "DELAPAN";
        break;
    case 9:
        $terbilang = "SEMBILAN";
        break;
    default:
        $terbilang = "Nilai di luar jangkauan";
        break;
}
echo "Bentuk Terbilang dari angka $angka adalah $terbilang";
?>
```

3

PHP Perulangan

1

```
<?php
for ($i = 0; $i < 10; $i++) {
    echo $i+1 . '<br />';
}
?>
```

2

```
<?php
$i = 0;
while ($i < 10) {
    echo $i+1 . '<br />';
    $i++;
}
?>
```

3

```
<?php
$i = 0;
do {
    echo $i+1 . '<br />';
    $i++;
}
while ($i < 10);
?>
```

```
<?php
$array = array('PHP', 'Python', 'Ruby');
foreach ($array as $value) {
    echo $value . '<br />';
}
?>
```

4

5

```
<?php
$array = array('Java', 'PHP', 'C++');
foreach ($array as $key => $value) {
    echo $key . ' yaitu Bagian dari ' . $value . '<br />';
}
?>
```

Any Question?